

Theoretical Development and Study of Takeoff Constraint Thrust Equation for A Drone

C. Huang Shen¹, F.Y.C. Albert², CK Ang³, Dwee Jin Teck⁴, K.P Chan⁵

^{1,2,4}UCSI University

Department of Electrical and Electronic Engineering, Cheras, 56000 Kuala Lumpur, Malaysia

Corresponding author: Chuahs@ucsiuniversity.edu.my, hschuabro@gmail.com

³UCSI University

Department of Mechanical Engineering, Cheras, 56000 Kuala Lumpur, Malaysia

⁵UCSI University

Department of Mechatronic Engineering, Cheras, 56000 Kuala Lumpur, Malaysia

Abstract—This paper presents theoretical development for a various weight of drone to take off away from land. This theory equation is important to consider when selecting a motor reflect to the total weight of drone must be able to comfortably to take off and fly. The theory is based on the momentum theory, disk actuator theory, kinetic energy and power of motor with propeller. The thrust must be twice than the flying weight of the drone. This theory equation results are useful for the drone application in the extra features design during the fly such as the camera, thermal image sensor, wireless sensor and other sensors application. The results are validated using a paid website calculator from “www.eCalc.ch”. The power factor is given by the website is 3.5 and was compared with the theory equation. The power factor from the theory equation must adjusted and tuned to 3.15 in order in line with the “eCalc” results of the graph plotting.

Keywords—Drone; Thrust estimation; Performance equation;

I. INTRODUCTION

Drone has become an age of no longer like a toys or playing fly for interested and enthusiast. Drone can be considered as an Unmanned Air Vehicle (UAV). The drone technology has matured enough providing many applications like thermography, agriculture [1], disaster management [2], military drone [3], Thermal observation [4], Disaster monitoring and management [5], state agency investigations, landslide and other steep terrain investigations, bridge inspections, roadside and roadway inspections [6] and others. A quadcopter is a short form for quadrotor helicopter which is commonly known as quadrotor and drone as well. This mechanism is using multirotor to be lifted and propelled against the gravity with four rotors that widely used nowadays for many purposes. A pair diagonal propeller will rotate clockwise and another pair will rotate counter-clockwise. These rotation speed of a motor is used to control the direction and achieve the movement of the drone. The speed of each motor is directly proportional to the thrust as the rotational speed of the motor increase the force that thrust the drone will also increase and make the drone to fly. A quadrotor UAV is required Global Positioning System (GPS), Gyroscope, flight controller, transceiver, motor speed driver, flight controller, lipo-Battery, brushless DC motor, propeller and quadcopter airframe. The fundamental components set up for a drone are shown in fig.1. GPS was not required in this research project because of it was focused in the propeller thrust estimation. A

suitable brushless DC motor, type of battery and propeller apply to the particular drone design is very important, different DC motor and type of propeller and battery are going to affect the thrust to move the quadcopter airframe up from the ground. A minimum weight of thrush is needed in order to lift up the quadcopter. The total weight of quadcopter is based on the battery, flight controller, propeller, motor, driver, camera, frame, and receiver. The calculation must confirmed that the motor is suitable for the application. Ampere is one of the considerations to choose the motor, battery and propeller.

This paper is organised and divided into 4 sections as follows: Section II presents the theory analysis of the equilibrium propeller thrust. Section III presents the calculation of take-off thrust while Section IV results investigation under which of the parameter is in line or identical with a paid website calculation conditions. The figure 1 is shown the fundamental components setup for a drone system.

II. THEORY ANALYSIS OF THE EQUILIBRIUM PROPELLER THRUST

The theory of the propeller thrust is based on the Momentum theory [7], which is also known as helicopter rotor (disk actuator theory) The thrust force can be calculated as in the mathematic terms equation (1) [8, 9]:

$$T = A \times \rho \times v \times \Delta v \quad [1]$$

Where T represents the thrust in Newton [N], A represents the area of the propeller rotor in meter square (m²), ρ represents the density of air in kilogram per cubic meter (kg/m³), v represents

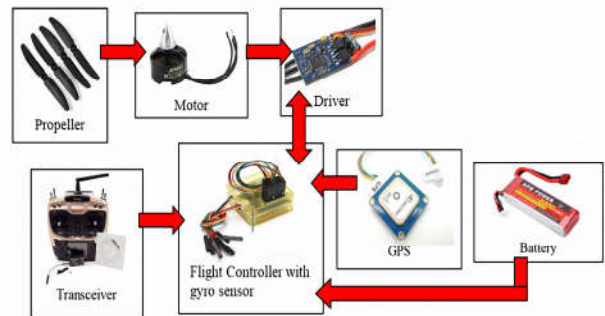


Fig. 1. Fundamental components setup

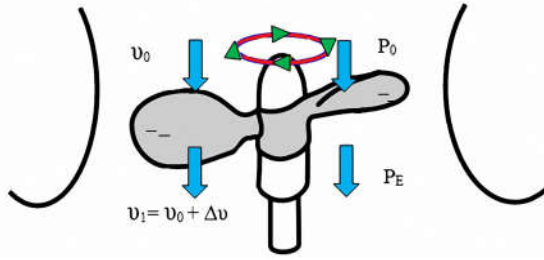


Fig. 2. Propeller air flow

the velocity of air at the propeller in meter per second (m/s), Δv represents the velocity of air accelerated by propeller in meter per second.

The equation 1 is needed to derive further more according to the type of propeller blade. Consider a full power of motor propeller air flow without the losses as shown in Fig. 2. In the air flow diagram above shown P_0 and P_E are the pressure difference before and after of the propeller. The equation (2) and (3) of the mass flow and the change of the air flow are derive as shown below:

$$\dot{m} = A \times \rho \times \Delta v \quad (2)$$

$$\text{Momentum conservation, } F = \dot{m} \times v \quad (3)$$

Where $\dot{m}$ represents the mass flow rate in kilogram per second [kg/s], ρ represents the density of air in kilogram per cubic metre (kg/m^3), v represents the velocity of air at the propeller in meter per second (m/s), Δv represents the velocity of air accelerated by propeller in meter per second. The conservation of energy is given by:

$$\text{Energy spent in one act} = \text{Energy gained in the related act} \quad (4)$$

Similarly, the propeller spinning energy available in the air flow is converted to force against the gravity which created the thrust. Suppose a velocity of propeller motor is spanned (rpm) for a particular velocity air accelerated

The conservation of energy equation can be written as

$$\text{Energy conservation, } F \times \Delta v = \frac{1}{2} \times \dot{m} \times v^2 \quad (6)$$

Substituting equation (2) and (3) into equation (6)

$$\dot{m} \times v \times \Delta v = \frac{1}{2} \times A \times \rho \times \Delta v \times v^2 \quad (7)$$

Substituting equation (2) into equation (7), and simplify.

$$A \times \rho \times \Delta v \times v \times \Delta v = \frac{1}{2} \times A \times \rho \times \Delta v \times v^2 \quad (8)$$

$$v = 2 \times \Delta v \quad (9)$$

Substituting equation (9) into equation (1)

$$T = 2 \times A \times \rho \times \Delta v^2 \quad (10)$$

Refer to the Fig. 2. , the area of the rotor diameter is:

$$A = \pi \times R^2 \quad (11)$$

Where R represents the propeller radius in meter [m].

Substituting equation (11) into equation (10)

$$T = 2 \times \pi \times R^2 \times \rho \times \Delta v^2 \quad (12)$$

From the equation 12, the Δv , velocity of air accelerated by propeller is produce by the force and velocity which can be

$$\text{derived from the equation of } \Delta v = \frac{P}{T}$$

Where P represents the power of the motor transmitted to propeller in watt [W], T represents the thrust in newton [N]. Substituting Δv into equation (12)

$$T = (2 \times \pi \times R^2 \times \rho \times P^2)^{\frac{1}{3}} \quad (13)$$

The power is calculated based on the motor run in load condition RPM (Revolutions per minute). Normally, the specification is given in no load condition of the RPM per volt. The rpm value should be calculated 75% approximately of its no load rpm. The power equation is based on the propeller constant and power factor for the propellers that are used to create thrust [10]. The power factor was tested according to the propellers constant from Mr. P. Connolly for Hyperion (APC propellers).

$$P = \text{Propeller constant} \times \text{rpm}^{\text{power factor}} \quad (14)$$

The Propeller constant is calculate by P/D ratios. D is the diameter length of a propeller of N blades while P is the pitch of the blade. Power factor is the ratio of operation power in kilowatts (kW) to apparent total power in kilovolt-amperes (kVA)

The RPM is calculate based on supplied voltage to the motor as shown in equation 15. As the voltage increases, the spins is increases simultaneously.

$$\text{RPM} = K_v \times \text{Voltage} \quad (15)$$

Where K_v represents the RPM per volt from the specification of the motor. Voltage represents the supplied voltage from the battery to the motor.

Substituting equation (14) and (15) into equation (13), the completed equilibrium propeller thrust equation can be presented as in equation (16)

III. CALCULATION OF TAKEOFF THRUST

The procedures to obtain the takeoff thrust performances are

Fig. 3. A eCalc thrust calculator

$$T = (2 \times \pi \times R^2 \times \rho \times (\text{propeller constant} \times (\text{rpm})^{\text{power factor}})^2)^{\frac{1}{3}} \quad (16)$$

made an assumption of thrust without frictional losses. The theory calculation was validated using a paid website calculator from “www.eCalc.ch” as shown in figure 3. According to the equation 16, the parameters and coefficients were needed to find the thrust are propeller radius(R), pi (π), air density (ρ), APC propeller constant and power factor. The estimated propeller rpm is used to calculate the thrust. The APC propeller speeds are controlled by high power Kv motors, the vehicle motion can be neglected affect the motor speed. The power factor and propellers constant are based on the aircraft world from Mr. P. Connolly for Hyperion (APC propellers) [10].

A. Parameters and coefficient value for performance calculation

To compare and analyse the calculation results established in equation 16 and eCalc, this section is define all the values of propeller radius(R), pi (π), air density (ρ), APC propeller constant and power factor and rpm.

IV. RESULTS AND ANKYSIS

Using the eCalc thrust calculator and the equations 16 for equilibrium propeller thrust can be estimated for a given set

of parameters in table 1. Fig. 4 shows the theoretical calculation comparison with the simulation results.

For an example let the rpm is equal to 10000 rpm.

$$T = (2 \times \pi \times 0.1143^2 \times 1.225 \times (0.048 \times 10^{3.55})^2)^{\frac{1}{3}} = 14.287 N$$

TABLE I. PARAMETERS AND COEFFICIENT VALUES

Parameters	Values
Propeller modal	APC slow fly series(Prop. Size 9*38)
Propeller radius(R),	0.1143 m
Pi (π),	3.14159
Air density (ρ),	1.225 Kg/m ³
APC propeller constant	0.048
power factor	3.55
rpm	3000 rpm to 11000 rpm

The comparison of the thrust from the eCalc simulation and theory calculation is shown in Table 2 for the motor speeds from 3,500 rpm to 11,300 rpm. The relative error, designated

$$E_T = \left(\frac{\text{Theory} - \text{Simulation}}{\text{Theory}} \right) \times 100\% \quad (17)$$

TABLE II. COMPARISON THRUST MOTOR AND RELATIVE ERROR WITH THE POWER FACTOR 3.55

RPM	Thrust Motor		Relative errors %
	Simulation (P.F=3.55)	Theoretical (P.F=3.55)	
3586	1	1.261	21%
4811	1.8	1.593	-13%
5072	2	2.865	30%
5783	2.6	3.909	33%
6991	3.8	6.12	37%
7857	4.8	8.073	41%
8930	6.2	10.93	43%
9073	6.4	11.349	43%
9887	7.6	13.9	45%
10016	7.8	14.341	46%
11341	10	19.244	48%

According to the relative error results, The table did not show that a constant data percent variation on the estimation of 75% efficiency performance. The power factor of the theoretical was recalculated again using the values of 3.3, 3.2, 3.15 and 3.1 as shown in fig. 5.

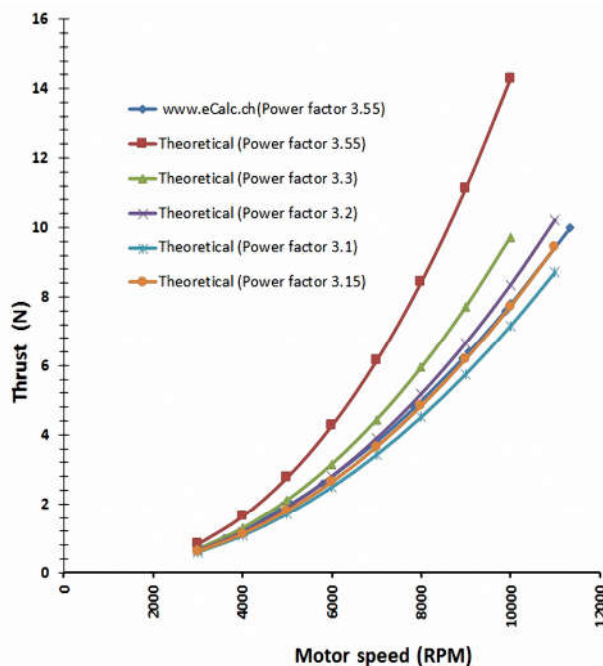


Fig. 5. Relationship between the theoretical and calculator motor thrust with different power factor

A graph plot result is statistically significant when the power factor is 3.15. The relative errors were recalculated again and obtained a close result between the simulation and theoretical as shown in Table 3.

TABLE III. COMPARISON THRUST MOTOR AND RELATIVE ERROR WITH THE POWER FACTOR 3.15

RPM	Thrust Motor		Relative errors %
	Simulation (P.F=3.15)	Theoretical (P.F=3.15)	
3586	1	0.89	-12.00%
4811	1.8	1.05	-17.00%
5072	2	1.86	-0.14%
5783	2.6	2.45	-0.15%
6991	3.8	3.65	-0.15%
7857	4.8	4.66	0.14%
8930	6.2	6.1	-0.10%
9073	6.4	6.3	-0.10%
9887	7.6	7.55	-0.05%
10016	7.8	7.76	0.05%
11341	10	10.07	0.07%

V. CONCLUSION

As conclusion, the Quadcopter motor rpm theoretically and simulation comparison result obtained. The study on the power factor reflected to the motor speed of the output thrust are approximately identical to the simulation results from the software specification used to calculate the thrust. So, in this study, the theoretical calculation power factor was fine-tuned to match the simulation results. After the calculation, the power factor results obtained was 3.15. In future work, the experiments should be carried out to validate the theoretical and simulation results.

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